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Analyzing and Predicting Hotel Reservation Cancellations Using Logistic Regression: Executive Summary

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Statement of the Problem and the Hypothesis

The cancellation of hotel reservations represents a significant challenge for the hotel industry, as it can lead to an inefficient allocation of available rooms. Additionally, it can cause difficulties in predicting hotel revenue due to the unforeseen likelihood that a hotel reservation will be cancelled prior to the guest's potential stay.

Mateer (2024) highlights the problem facing the hotel industry by stating that "The hotel industry's current model, which allows guests to cancel up to 24 hours before their arrival, leaves hoteliers in a precarious position, vulnerable to last-minute cancellations and exposed to unexpected revenue loss and vacancies." This sentiment reinforces the value that can be obtained by the hotel by being able to better identify the type of potential guest that is more likely or less likely to cancel their reservation.

The primary goal of this data analysis process was to determine whether factors could be identified that significantly influenced the likelihood that a hotel reservation would be canceled. Additionally, the data analysis sought to determine the likelihood that any single reservation would be canceled given a set of identifying factors. The null hypothesis for this analysis was that no identified variables had a significant influence on the likelihood that a hotel reservation was canceled. The alternative hypothesis was that at least one of the identified independent variables had a significant influence on the likelihood that a hotel reservation was canceled.

Summary of the Data-Analysis Process

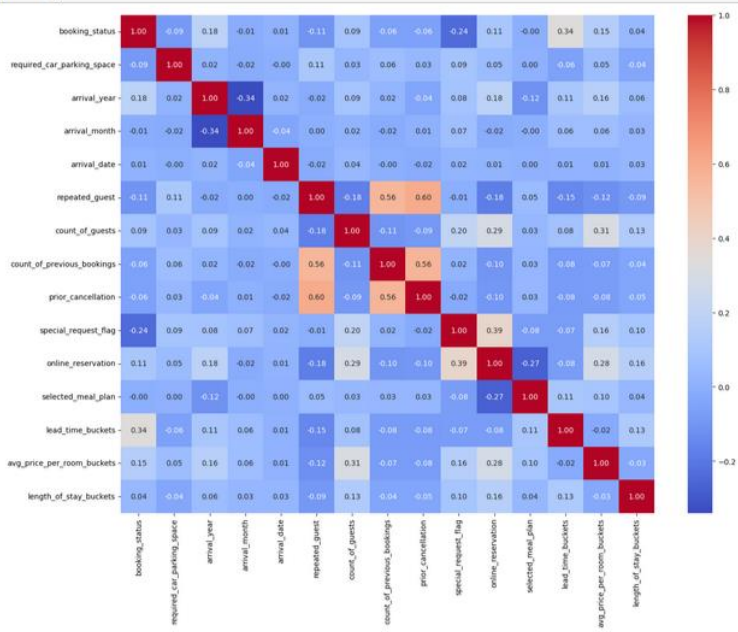
The first step in the data-analysis process was to source the necessary data from Kaggle.com titled "Hotel Reservations Dataset" (Raza, 2023). This dataset contained 19 initial columns with a total of 36,275 observations.

Once the data was read into the programming environment, exploratory data analysis was performed. Next, checks were performed for duplicate and missing values. In this case, no duplicate or missing values were identified. The distinct values for quantitative variables were identified in order to determine whether any variables needed to be modified for increased usefulness in the later logistic regression model. Additionally, columns were combined to create

new columns that were easier to interpret. New variables (such as high value and low value buckets) were created in order to more easily use those variables as explanatory variables in the logistic regression model. Categorical variable re-expression was performed so that the categorical variables could be featured in the regression model as well. Outlier checks were performed using Z-scores and histograms, however no outliers requiring treatment were identified. Bivariate visualizations were performed using the target variable of whether the reservation was canceled as one of the variables and potential explanatory variables as the other variable.

A key aspect of the data-analysis process was to handle multicollinearity. Murel (2024) explains the problem of multicollinearity by stating, “Collinearity denotes when two independent variables in a regression analysis are themselves correlated; multicollinearity signifies when more than two independent variables are correlated. Their opposite is orthogonality, which designates when independent variables are not correlated. Multicollinearity prevents predictive models from producing accurate predictions by increasing model complexity and overfitting.” In order to ensure that the model is able to produce accurate predictions, multicollinearity was handled using a combination of correlation heatmaps and Variance Inflation Factor.

```
In [98]: hotel_data_for_corr = hotel_data[['booking_status', 'required_car_parking_space','arrival_year', 'arrival_month', 'arrival_
                                           'count_of_guests', 'count_of_previous_bookings', 'prior_cancellation', 'special_request_flag',
                                           'online_reservation', 'selected_meal_plan', 'lead_time_buckets', 'avg_price_per_room_buckets',
                                           'length_of_stay_buckets']]
         corr_matrix = hotel_data_for_corr.corr()
         plt.figure(figsize=(16, 12))
         sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
         plt.show()
```



```
In [99]: #Feature selection via VIF
         hotel_data_vif = hotel_data_for_corr.drop(['booking_status'], axis=1)
         print(hotel_data_vif.columns)

         vif_data = pd.DataFrame()
         vif_data["feature"] = hotel_data_vif.columns

         vif_data["VIF"] = [variance_inflation_factor(hotel_data_vif.values, i)
         for i in range(len(hotel_data_vif.columns))]
         print(vif_data)

Index(['required_car_parking_space', 'arrival_year', 'arrival_month',
       'arrival_date', 'repeated_guest', 'count_of_guests',
       'count_of_previous_bookings', 'prior_cancellation',
       'special_request_flag', 'online_reservation', 'selected_meal_plan',
       'lead_time_buckets', 'avg_price_per_room_buckets',
       'length_of_stay_buckets'],
      dtype='object')
                      feature        VIF
0   required_car_parking_space   1.063676
1                 arrival_year  27.468311
2                arrival_month   6.989041
3                 arrival_date   4.203395
4               repeated_guest   1.914826
5              count_of_guests  12.180620
6   count_of_previous_bookings   1.670996
7           prior_cancellation   1.797029
8         special_request_flag   2.249881
9           online_reservation   4.188630
10          selected_meal_plan   8.102367
11           lead_time_buckets   2.143051
12  avg_price_per_room_buckets   2.222835
13      length_of_stay_buckets   1.555042
```

The above figures demonstrate the two-pronged approach to identifying variables that are highly correlated with each other. In this data-analysis process, a cutoff of a VIF value of 5 was used to determine the threshold above which feature reduction would be performed. This cutoff value is justified by Singh (2024), who states that "VIF > 5: High multicollinearity is present. The predictor's standard error may be noticeably inflated, which can make its coefficient less reliable. Consider taking steps to reduce multicollinearity, like removing or combining correlated predictors."

Once multicollinearity was sufficiently addressed, an initial logistic regression model was constructed. Backward stepwise elimination was performed until all remaining independent variables were statistically significant at a 95% confidence level. After removing four statistically insignificant variables via the backward stepwise elimination process, the final logistic regression results were obtained.

```
In [105]: ##Remove least significant variable and regress again (arrival_date)
          independent_variables = ['required_car_parking_space', 'repeated_guest',
                                   'special_request_flag', 'online_reservation', 'lead_time_buckets',
                                   'avg_price_per_room_buckets']
          X = hotel_data[independent_variables]
          y = hotel_data['booking_status']

          X = sm.add_constant(X)
          final_model = sm.Logit(y, X)
          final_result = final_model.fit()
          print(final_result.summary())
```

```
Optimization terminated successfully.
         Current function value: 0.495038
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:         booking_status   No. Observations:                36275
Model:                          Logit   Df Residuals:                    36268
Method:                           MLE   Df Model:                            6
Date:                Wed, 08 Jan 2025   Pseudo R-squ.:                  0.2173
Time:                        12:00:43   Log-Likelihood:                -17958.
converged:                       True   LL-Null:                       -22944.
Covariance Type:            nonrobust   LLR p-value:                     0.000
=================================================================================================
                                    coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------------------
const                            -2.1864      0.033    -66.706      0.000      -2.251      -2.122
required_car_parking_space       -1.4541      0.111    -13.131      0.000      -1.671      -1.237
repeated_guest                   -1.8772      0.260     -7.231      0.000      -2.386      -1.368
special_request_flag             -1.7919      0.031    -58.294      0.000      -1.852      -1.732
online_reservation                1.2764      0.032     40.114      0.000       1.214       1.339
lead_time_buckets                 1.7715      0.028     62.546      0.000       1.716       1.827
avg_price_per_room_buckets        0.8414      0.027     30.819      0.000       0.788       0.895
=================================================================================================
```

Outline of the Findings

The findings of the data analysis process are demonstrated in part by the final logistic regression model results above. Based on the model results, we were able to derive the final logistic regression formula, which is as follows:

*log-odds(P(booking_status=1))= −2.1864 − 1.4541*required_car_parking_space − 1.8772*repeated_guest − 1.7919*special_request_flag + 1.2764*online_reservation + 1.7715*lead_time_buckets + 0.8414*avg_price_per_room_buckets.*

This formula can be used to determine the log-odds that any individual reservation will be canceled given the set of criteria determined in the right side of the formula. At 6% likelihood of cancellation, the scenario in which a reservation is least likely to be canceled is one where a parking space is required, the guest is a repeated guest, the guest has a special request, the reservation is not an online reservation, there is a short lead time, and the room has a low price. At 84% likelihood of cancellation, the scenario in which a reservation is most likely to be canceled is one where a parking space is not required, the guest is not a repeat guest, the guest

does not have a special request, the reservation is an online reservation, there is a long lead time, and the room has a high price.

Overall, the results of the data-analysis provide a framework for determining the likelihood that a hotel reservation will be canceled. From the log-odds formula, it can be seen that requiring a parking space, being a repeat guest, and having a special request lower the likelihood that the reservation will be canceled. Intrinsically, this supports the argument that the more invested a guest is in the reservation, the lower the likelihood that the reservation will be canceled. Requiring a parking space could mean that the guest is also renting a car, which would mean that any hotel cancellation would be accompanied by a rental car reservation cancellation, increasing the needed friction for any cancellation to be performed. A repeat guest has already visited the hotel successfully, so it would logically follow that a guest that has visited the hotel prior without canceling would be less likely to cancel a future reservation. Finally, having a special request could represent more of an investment of thought in the reservation, which would likely mean that a reservation that had more thought put into it would have a lower likelihood of cancellation.

In the inverse, booking the reservation online, booking the reservation relatively far out from the stay in terms of time, and the room being a relatively high cost all increase the likelihood that the reservation is canceled. Each of these factors have logical explanations, as the higher than average lead time means that there are a greater than average number of days in which an outside event could occur that would require the reservation to be canceled. Booking the reservation online represents a lower initial investment than booking by alternate means, as an online reservation can be more of a spur of the moment decision than booking the reservation in person. Finally, a higher cost room represents a greater monetary value from cancellation than a lower cost room, as cancelling the reservation would return more money to the potential guest.

```
In [107]: #Calculate accuracy score
          TN = 22792
          FP = 1598
          FN = 7212
          TP = 4673

          accuracy_score = (TP+TN)/(TP + TN + FP + FN)
          print(accuracy_score)

          0.7571330117160578
```

In the above figure, we can see that the developed logistic regression model had an accuracy score of approximately 75.7%. This score is a decent result for the model, which implies that this logistic regression model is a good starting point that can be further refined with a variety of potential methods.

Explanation of the Limitations of the Techniques and Tools Used

The primary technique used to perform this data analysis was logistic regression, which can have a host of limitations. One such limitation occurs due to the imbalance of the dataset used to perform the data analysis.

```
In [23]: print(hotel_data.booking_status.value_counts())

booking_status
0    24390
1    11885
Name: count, dtype: int64
```

Above, we can see that the target variable of booking status has significantly more instances of a reservation not being canceled than a reservation being canceled. GeeksforGeeks (2024) highlights the impact of this issue by stating, “When faced with imbalanced datasets, the model tends to be biased towards the majority class. As a result, it may have lower sensitivity (true positive rate) for the minority class, leading to poor performance in predicting rare events.”

In addition to the limitation of the logistic regression model, the tool of backward stepwise elimination also comes with limitations. Hayes (2022) notes that “Statisticians note several drawbacks to the approach, including incorrect results, an inherent bias in the process itself, and the necessity for significant computing power to develop complex regression models through iteration.” This limitation presents primarily in the form of being a resource-intensive process. While this was not an issue in this data analysis due to the relatively few amount of eliminations needed, in a process that required more individual eliminations, the backward stepwise elimination process could quickly cause issues with requiring computing power.

Summary of Proposed Actions

The primary course of action that would be proposed based on the results of the data analysis would be to use the model on additional data so that model performance can be assessed outside the context of the original data. The current accuracy score of the model indicates that it could potentially be beneficial on additional data, however the model would need to be used on that additional data to determine whether those benefits present themselves.

A secondary course of action would be to use different metrics for accuracy to assess the current logistic regression model. As noted previously, the imbalance in the target variable indicates that the model could have issues with sensitivity. Once a variety of different metrics are used, it could then be determined whether the model performs differently on a sample of the data where the target variable is not imbalanced. For example, a sample of 10,000 reservations that were canceled and 10,000 reservations that were not canceled could be obtained so that the model could be run and analyzed on a balanced dataset.

A third proposed action would be to interact potential independent variables with each other in order to potentially identify relevant information that is currently hidden due to the lack of interacted variables in the current logistic regression model. While the current model is relatively effective, the model's value could be increased by including newly derived variables that could increase the explanatory power of the logistic regression model.

Expected Benefits of the Study

As-is, a study of the performed data analysis could have a variety of benefits for a hotel. One such benefit is that the logistic regression model created as a result of the data analysis can be used to determine the likelihood that any single reservation would be canceled. Above, it was determined that the likelihood of cancelation ranged from approximately 6% at the low end to 84% at the high end. Additionally, a variety of combinations of independent variable outcomes can be plugged into the model to assess likelihoods of cancellation in between 6% and 84%. Based on this, the hotel can implement policies designed to limit potential cancellations, as well as provide enhanced customer service to the reservations that are least likely to be canceled.

For example, reservations identified to be at the highest risk of cancellation could benefit from a reminder system where potential guests are asked ahead of time to confirm their reservations. This would benefit the hotel as it would allow them to have more time to offer the room to another potential guest.

In addition to handling guests most likely to cancel, hotels could also benefit by providing enhanced customer service to the reservations that are the least likely to be cancelled. Providing enhanced service to all reservations comes at an increased risk to the hotel, as the increased quality of service could be provided to a reservation at high risk of being canceled. However, by limiting this service only to the reservations most likely to be converted to guest visits would allow the hotel to efficiently provide elevated quality of service to reservations most likely to be in a position to appreciate that service.

Finally, a major benefit of this data analysis study is that it can provide a foundation for improved research going forward. By identifying the initial value of such a study, further studies can head down the same path and provide more valuable and specific benefits to a hotel.

References

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